



SCHOOL OF COMPUTING PROJECT GUIDELINE

(Computer Science, Cyber Security & Software Engineering)

Student Guide for Final Year Projects

INTRODUCTION

This Guideline is developed to guide final-year students in BSc Computer Science, BSc Cyber Security, and BSc Software Engineering on how to plan, write, and present their final year projects. It is adapted from the standard project Table and aligned with NUC Core Curriculum and Minimum Academic Standards (CCMAS).

The final year project is a compulsory academic requirement and represents the peak of undergraduate training. It tests a student's ability to apply knowledge, conduct independent work, and solve real-world computing problems.

PROGRAMME SPECIFIC EMPHASIS

BSc Computer Science

- Algorithm Design and Analysis
- Computational Efficiency
- Data Structures
- Sometimes Software and Application Development
- Sometimes Network Analysis and Design

BSc Cyber Security

- Threat Modeling and Risk Assessment
- Security Architecture
- Vulnerability Assessment and Penetration Testing
- Network Security Analysis and Design

BSc Software Engineering

- Requirements Engineering
- SDLC Models
- Software Testing and Maintenance

- Software and Application Development

STRUCTURE OF THE FINAL YEAR PROJECT (NUC FORMAT)

According to the NUC Core Curriculum and Minimum Academic Standards (CCMAS), the Final Year Project is divided into two distinct courses carried out over two semesters:

- Project I – First Semester
- Project II – Second Semester

Both courses are compulsory and must be completed sequentially. Project I focuses on proposal development and design, while Project II focuses on implementation, testing, and evaluation.

PROJECT I – FIRST SEMESTER

(Proposal, Literature Review and System Design)

Project I is research- and design-oriented. It assesses the student's ability to define a problem, review existing works, and propose a technically sound solution.

BSc Computer Science

CSC 497: Final Year Project I (3 Units C: PH 135)

Learning Outcomes: At the end of this course, students should be able to:

1. Identify a researchable project topic in Computer Science.
2. Search and review literature pertinent to the identified problem statement.
3. Acknowledge and reference sources of information used in the research report.
4. Conceptualize and design a research methodology to address an identified problem.
5. Determine tools for analyzing data collected based on research objectives.
6. Write a coherent proposal on the research project to be conducted.
7. Orally present the written project proposal.

Course Contents:

- Independent or group investigation of appropriate software, hardware, communication, network, or IT-related problems in Computer Science under the supervision of a lecturer.

- Before registering, students must submit a written proposal outlining the project, estimated schedule of completion, and computer resources needed.
- A formal written report is essential; an oral presentation may also be required.

BSc Cybersecurity

CYB 497: Final Year Project I (3 Units C: PH 135)

Learning Outcomes: At the end of this course, students should be able to:

1. Identify a researchable project topic in Cybersecurity and work on the introductory aspect under the guidance of the supervisor.
2. Review relevant recent literature pertinent to the problem statement and establish the research gap.
3. Acknowledge and reference sources of information used in the research report.
4. Conceptualize and design a research methodology to address an identified problem.
5. Determine tools for analyzing data collected based on research objectives.
6. Write a coherent proposal on the research project to be conducted.
7. Orally present the written project proposal.

Course Contents:

- Independent or group investigation of cybersecurity-related problems under lecturer supervision.
- Written proposal must include statement of problem, aim, objectives, scope, significance, research gap, proposed methodology, schedule, and resources needed.
- Formal written report is essential; oral presentation may also be required.
- Topics may include AI applications, steganography, quantum computing, big data, cloud security, ethical hacking, cyber hunting, penetration testing, network intrusion and prevention, threat management, biometrics, digital image processing, Blockchain, IoT, 5G security, etc.

BSc Software Engineering

SEN 497: Final Year Project I (3 Units C: PH 135)

Learning Outcomes: At the end of this course, students should be able to:

1. Identify researchable project topics in Software Engineering.
2. Search and review literature pertinent to identify problem statements.

3. Acknowledge and reference sources of information used in the research report.
4. Conceptualize and design a research methodology to address an identified problem.
5. Determine tools for analyzing data collected based on research objectives.
6. Write a coherent report on the research conducted.
7. Take instruction to accomplish the set goals for the project under the guidance of the research supervisor.
8. Orally present the written project report.

Course Contents:

- Independent or group investigation to address a Software Engineering problem under lecturer supervision.
- Written proposal must include project outline, estimated schedule, and computer resources.
- Formal written report is essential; oral presentation may also be required.
- At the end of the semester, submission of introduction, literature review, and methodology for grading.

FORMAT FOR PROJECT I WRITE-UP**Preliminary Pages**

- Title Page
- Certification Page
- Abstract
- Table of Contents
- List of Tables
- List of Figures
- List of Abbreviations

Chapter One: Introduction

- Background of the Study
- Statement of the Problem
- Research Questions (if applicable)
- Aim and Objectives of the Study
- Scope and Limitations
- Significance of the Study

- Definition of Terms
- Project Organization

Chapter Two: Literature Review

- Introduction
- Conceptual Review
- Theoretical Framework
- Review of Related Works
- Summary and Research Gap

Chapter Three: System Analysis and Design

- Introduction
- Research Methodology
- Existing System Analysis
- Problem of the Existing System
- Proposed System Description
- System Requirements (Hardware & Software)
- System Design (Architectural, Database, Input and Output or User Interface Design)
- System Architecture and UML Diagrams

BSc COMPUTER SCIENCE PROJECT I

(Use the general format, with emphasis on algorithms and data processing)

Chapter Three (Special Focus):

- Algorithm Design and Analysis
- Data Structures Used
- Computational Complexity (if applicable)

BSc CYBER SECURITY PROJECT I

(Security-oriented emphasis)

Chapter Two (Add):

- Security Concepts and Models
- Threats, Vulnerabilities and Attacks

- Security Standards and Frameworks

Chapter Three (Modify):

- Security Requirements Analysis
- Risk Assessment Methodology
- Security Architecture Design
- Cryptographic Techniques (if applicable)

BSc SOFTWARE ENGINEERING PROJECT I

(Strong focus on SDLC and engineering practices)

Chapter Two (Add):

- Software Development Life Cycle (SDLC) Models
- Software Engineering Principles

Chapter Three (Modify):

- Requirements Engineering
- System Modeling (UML Diagrams)
- Software Design Patterns
- Version Control and Project Management Tools

ASSESSMENT FOR PROJECT I (FIRST SEMESTER)

Project I is assessed mainly on research quality and proposal strength.

Typical assessment components include:

Assessment Component	Weightage
Approved Project Proposal	20%
Quality of Literature Review	30%
System Analysis and Design	30%
Proposal Seminar / Presentation	20%
Total	100%

Project I – Proposal Seminar/Presentation Rubric

Criteria	Description	Score (%)
Problem Definition	Clarity and relevance of problem statement	20
Literature Review	Depth, relevance, and citation quality	25
Methodology & Design	Feasibility and technical soundness	25
Presentation Skills	Clarity, confidence, and organization	15
Response to Questions	Understanding and justification	15
Total		100 (convert to 20%)

PROJECT II – SECOND SEMESTER

(Implementation, Testing, Documentation and Defence)

Project II is implementation and evaluation-oriented. It assesses the student's ability to build, test, and defend the proposed system.

BSc Computer Science

CSC 498: Final Year Project II (3 Units C: PH 135)

Learning Outcomes: At the end of the course, students should be able to:

1. Demonstrate technical skills in Computer Science.
2. Demonstrate generic transferable skills such as communication and teamwork.
3. Produce a technical report for the chosen project.
4. Defend the written project report.
5. Appreciate the art of carrying out full-fledged research.

Course Contents:

- Continuation of CSC 497, focusing on project implementation and evaluation.
- Chapters 4-5 must be approved by the supervisor; the final report (chapters 1-5) will be submitted for grading.
- An oral presentation is required.

BSc Cybersecurity

CYB 498: Final Year Project II (3 Units C: PH 135)

Learning Outcomes: Upon completion of the project, students should be able to:

1. Conduct research methodology, analyse collected data, document the project report using the recommended format and referencing style.
2. Develop the prototype or simulated model of research study with programming, if required.
3. Produce the report.
4. Prepare PowerPoint presentation and defend the project in the final oral examination with an external examiner.

Course Contents:

- Continuation of CYB 497, including research methodology, data analysis using statistical tools, performance evaluation, documentation, programming of prototype/simulation, plagiarism and grammar check, and PowerPoint presentation skills.
- Chapters 4-5 approved by the supervisor; final report submitted for grading.
- Oral presentation is required.

BSc Software Engineering**SEN 498: Final Year Project II (3 Units C: PH 135)**

Learning Outcomes: Upon completion of the project, students should be able to:

1. Demonstrate technical skills in Software Engineering.
2. Demonstrate generic transferable skills such as communication and teamwork.
3. Produce a technical report for the chosen project.
4. Defend the written project report.
5. Appreciate the art of carrying out full-fledged research.

Course Contents:

- Continuation of SEN 497, focusing on project implementation and evaluation.
- Chapters 4-5 must be approved by the supervisor; final report (chapters 1-5) submitted for grading.
- Oral presentation is required.

FORMAT FOR PROJECT II WRITE-UP

Preliminary Pages

- Title Page
- Certification Page
- Declaration
- Dedication (Optional)
- Acknowledgements
- Abstract
- Table of Contents
- List of Tables
- List of Figures
- List of Abbreviations

Chapter Four: Implementation, Testing and Evaluation

- Implementation Procedures
- Testing Strategies
- Test Cases and Results
- Performance Evaluation
- Security Evaluation (mandatory for Cyber Security projects)
- Discussion of Results

Chapter Five: Summary, Conclusion and Recommendations

- Summary of the Study
- Achievements of the Project
- Limitations of the Study
- Conclusion
- Recommendations

BSc COMPUTER SCIENCE PROJECT II

(Use the general format, with emphasis on algorithms and data processing)

Chapter Four (Special Focus):

- Performance Evaluation
- System Efficiency and Accuracy

BSc CYBER SECURITY PROJECT II

(Security-oriented emphasis)

Chapter Four (Add):

- Security Testing and Penetration Testing
- Vulnerability Assessment Results

BSc SOFTWARE ENGINEERING PROJECT II

(Strong focus on SDLC and engineering practices)

Chapter Four (Add):

- Software Testing Methods
- User Acceptance Testing
- Deployment and Maintenance Plan

ASSESSMENT FOR PROJECT II (SECOND SEMESTER)

Project II is assessed mainly on technical depth, implementation quality, and defence.

Typical assessment components include:

Assessment Component	Weightage
System Implementation and Functionality	30%
Testing and Evaluation	20%
Project Report (Documentation Quality)	20%
Project Demonstration	10%
Oral Defense	20%
Total	100%

Project II – Final Oral Defense Rubric

Criteria	Description	Score (%)
System Implementation	Functionality and originality	30
Testing & Evaluation	Correctness and performance analysis	20
Documentation	Organization, clarity, and formatting	15
Demonstration	Practical operation of the system	15

Criteria	Description	Score (%)
Oral Defense	Technical understanding and responses	20
Total		100 (Convert to 20%)

DETAILS OF THE PROJECT CONTENT

Preliminary Pages

Title Page

The first page of the report, including the project title, author's name, department and institution, degree program (e.g., BSc Computer Science, Cybersecurity, Software Engineering), and date of submission.

Certification Page

A page where the project supervisor and/or department head certifies that the project was carried out by the student under supervision and meets the required standard.

Declaration

A statement by the student declaring that the work is original and has not been submitted elsewhere, usually signed and dated by the student.

Dedication (Optional)

An optional page where the student dedicates the work to someone significant, such as family, mentors, or friends.

Acknowledgements

A page where the student expresses gratitude to individuals, organizations, or institutions that supported the research, including supervisors, lecturers, colleagues, and funding bodies.

Abstract

A concise summary of the project, typically 150–300 words, highlighting the purpose of the study, methods used, main findings or results, and conclusions or contributions.

Table of Contents

Lists all chapters, sections, and subsections with their corresponding page numbers to help readers navigate the report easily.

List of Tables

A numbered list of all tables included in the report, showing table titles and page numbers.

List of Figures

A numbered list of all figures, diagrams, and illustrations used in the report, including titles and page numbers.

List of Abbreviations

An alphabetical list of all acronyms and abbreviations used in the report, along with their full meanings for clarity.

Chapter One: Introduction

1.1 Background of the Study

Explains the context of the research topic, why it is important, and provides general information about the field or problem area. It sets the stage for the study.

1.2 Statement of the Problem

Clearly identifies the specific problem or issue the study aims to address. It highlights gaps, challenges, or deficiencies in existing systems or knowledge.

1.3 Research Questions (if applicable)

Lists the specific questions the research seeks to answer. These guide the investigation and help focus the study.

1.4 Aim and Objectives of the Study

- **1.4.1 Aim:** The general purpose or goal of the research.
- **1.4.2 Objectives:** Specific steps or milestones that need to be achieved to fulfill the aim.

1.5 Scope and Limitations

- **1.5.1 Scope:** Defines the boundaries of the study—what will and will not be covered.
- **1.5.2 Limitations:** Factors that may restrict the study or affect results (e.g., time, resources, data availability).

1.6 Significance of the Study

Explains the importance or contribution of the research. Shows how the findings will benefit individuals, organizations, or the field of study.

1.7 Definition of Terms

Clarifies key terms, acronyms, or jargon used in the study to ensure readers understand their meaning in context.

1.8 Project Organization

Provides an overview of how the report is structured, briefly describing the content of each chapter.

Chapter Two: Literature Review

2.1 Introduction

Introduces the purpose of the literature review, explaining why reviewing previous work is necessary.

2.2 Conceptual Review

Discusses key concepts, ideas, and theories related to the research topic. Helps establish a foundation for understanding the study.

2.3 Theoretical Framework

Presents the theory or model that underpins the study, guiding the research design and analysis.

2.4 Review of Related Works

Summarizes previous research studies or projects related to the topic, highlighting methods, findings, and limitations.

2.5 Summary and Research Gap

Provides a concise summary of reviewed literature and identifies gaps or areas where the current study will contribute new knowledge or solutions.

Chapter Three: System Analysis and Design

3.1 Introduction

Introduces the focus of the chapter, explaining the purpose of analyzing and designing the system.

3.2 Research Methodology

Describes the methods and approaches used to conduct the study (e.g., qualitative, quantitative, experimental, or survey methods).

3.3 Existing System Analysis

Examines the current system or process, highlighting how it works and any deficiencies.

3.4 Problem of the Existing System

Identifies challenges, inefficiencies, or limitations in the current system that the proposed system aims to solve.

3.5 Proposed System Description

Describes the new system being developed, including its features, functionalities, and intended benefits.

3.6 System Requirements (Hardware & Software)

Lists the necessary hardware, software, and tools needed to implement the proposed system.

3.7 System Design

- **3.7.1 Architectural Design:** Overall system structure and interaction of components.
- **3.7.2 Database Design:** Data structures, tables, and relationships.
- **3.7.3 Input and Output/User Interface Design:** How users interact with the system and how information is displayed or processed.

3.8 System Architecture and UML Diagrams

Visual representations of the system structure and behavior using diagrams such as Use Case, Class, Sequence, or Activity Diagrams.

Chapter Four: Implementation, Testing and Evaluation

4.1 Implementation Procedures

Describes step-by-step how the proposed system was developed and deployed. Includes installation, configuration, and integration of system components.

4.2 Testing Strategies

Outlines the approach used to test the system for correctness, functionality, and reliability. Can include unit testing, integration testing, system testing, and user acceptance testing.

4.3 Test Cases and Results

Presents the test cases designed to validate the system, including inputs, expected outputs, and actual results. Highlights whether the system meets the specified requirements.

4.4 Performance Evaluation

Assesses how well the system performs in terms of speed, efficiency, resource usage, and scalability.

4.5 Security Evaluation (*mandatory for Cybersecurity projects*)

Evaluates the system's security features, including vulnerability assessment, risk mitigation, authentication, and data protection measures.

4.6 Discussion of Results

Analyzes and interprets the results obtained from testing and evaluation. Compares expected outcomes with actual outcomes and explains any discrepancies.

Chapter Five: Summary, Conclusion and Recommendations

5.1 Summary of the Study

Provides a concise overview of the entire research, covering objectives, methodology, implementation, and findings.

5.2 Achievements of the Project

Highlights the key accomplishments and contributions of the project relative to its objectives.

5.3 Limitations of the Study

Discusses factors that constrained the study or affected the results, such as time, resources, or scope limitations.

5.4 Conclusion

Presents final remarks based on the study's findings, emphasizing the outcomes and their significance.

5.5 Recommendations

Suggests actions for future research, system improvements, or practical applications based on the project's results and limitations.

PROJECT REPORT FORMATTING GUIDELINES

1. Page Setup

Page Size: A4 (210 × 297 mm)

Margins:

Top: 2.5 cm

Bottom: 2.5 cm

Left: 3.5 cm

Right: 2.5 cm

Orientation: Portrait

Line Spacing: 1.5 lines for main text; single spacing for tables, figures, footnotes, and references

Paragraph Alignment: Justified

2. Font Style and Size

Main Text: Times New Roman, 12 pt

Headings:

Chapter Titles (e.g., **CHAPTER ONE: INTRODUCTION**): Times New Roman, 14 pt, Bold, Centered

Section Headings (e.g., 1.1 Background of the Study): Times New Roman, 12 pt, Bold, Left-aligned

Sub-section Headings (e.g., 1.1.1 ...): Times New Roman, 12 pt, Italic or Bold, Left-aligned

Tables and Figures Text: Times New Roman, 10–11 pt

Footnotes: Times New Roman, 10 pt

3. Pagination

Preliminary Pages (Title Page to List of Abbreviations): Roman numerals (i, ii, iii, ...)

Chapters (Introduction to Conclusion): Arabic numerals (1, 2, 3, ...)

Page Numbers Placement: Bottom center

4. Headings and Subheadings

Use a hierarchical numbering system for chapters, sections, and subsections:

Chapter: One, Two, Three ...

Section: 1.1, 1.2 ...

Subsection: 1.1.1, 1.1.2 ...

Headings should be **bold**; subheadings can be bold or italic.

5. Tables and Figures

Tables and figures must be numbered consecutively per chapter:

Example: Table 2.1 (first table in Chapter Two), Figure 3.2 (second figure in Chapter Three)

Provide captions above tables and below figures.

References to tables and figures in text should be explicit (e.g., “As shown in Table 3.1 ...”).

6. References and Citation

- Use consistent APA citation style.
- All cited works must appear in the Reference/Bibliography list at the end of the report.

7. Other Formatting Tips

- **Consistency:** Maintain consistent formatting throughout the report for fonts, headings, captions, and spacing
- **Plagiarism Check:** Ensure originality and proper referencing
- **Appendices:** Include supplementary materials, diagrams, code, or questionnaires in an appendix if necessary

WEEKLY PROJECT MILESTONES

Project I – First Semester Timeline (14 Weeks)

- **Week 1–2:** Topic selection and supervisor allocation
- **Week 3:** Proposal submission

- **Week 4:** Proposal approval and corrections
- **Week 5–7:** Literature review writing
- **Week 8:** Seminar presentation (Project Proposal Defence)
- **Week 9–11:** System analysis and design
- **Week 12:** Draft submission to supervisor
- **Week 13:** Corrections and revisions
- **Week 14:** Final submission of Project I report

Project II – Second Semester Timeline (14 Weeks)

- **Week 1–2:** Review of Project I and implementation planning
- **Week 3–6:** System development / configuration
- **Week 7:** Mid-semester progress review
- **Week 8–9:** System testing and debugging
- **Week 10:** Performance and security evaluation
- **Week 11:** Documentation and report writing
- **Week 12:** Final draft submission
- **Week 13:** Project demonstration and defence
- **Week 14:** Final bound submission